

# Technical specification

Valve-regulated lead-acid battery with Absorbent Glass Mat (AGM) separator. Sealed construction for standby power applications.

**12 V**

6 cells

**75 Ah**

10 h to 10.8 V at 25°C

**320 W/cell**

15 min to 1.67 V/cell at 25°C

## Electrical performance

|                       |                                 |
|-----------------------|---------------------------------|
| 10-hour capacity      | 75 Ah at 7.50 A to 10.8 V, 25°C |
| 20-hour capacity      | 80 Ah at 4.00 A to 10.8 V, 25°C |
| Internal resistance   | <=5.0 mOhms at 25°C             |
| Maximum discharge     | 670A for 5 seconds at 25°C      |
| Self-discharge        | 3% average per month at 20°C    |
| Design life / standby | 12 years                        |

## Charging and operating conditions

|                          |                                                            |
|--------------------------|------------------------------------------------------------|
| Charge method            | Constant voltage                                           |
| Float / cycle voltage    | 13.38-13.80 V / 14.40-14.70 V at 25°C                      |
| Maximum charge current   | 24A                                                        |
| Temperature compensation | Float: -20 mV/°C per battery; cycle: -30 mV/°C per battery |
| Operating temperatures   | Discharge: -20 to +60°C; charge: -10 to +60°C              |
| Storage temperature      | -20 to +60°C                                               |

## Mechanical specification

|                        |                                           |
|------------------------|-------------------------------------------|
| Dimensions (L × W × H) | 258 × 166 × 210 mm; overall height 215 mm |
| Approximate weight     | 24.5 kg ±5%                               |
| Terminal / torque      | See terminal drawing; Not specified       |
| Container              | ABS, UL94-HB                              |

# Construction & applications

High-output AGM construction for UPS standby power. Computer-designed grids, optimized paste and lead-calcium-tin plates support high energy density. Design life: 12+ years at 20°C.

## Battery construction

|                 |                   |
|-----------------|-------------------|
| Positive plate  | Lead dioxide      |
| Negative plate  | Lead              |
| Container / lid | ABS / ABS         |
| Safety valve    | Rubber            |
| Terminals       | Copper            |
| Separator       | Glass fibre (AGM) |
| Electrolyte     | Sulfuric acid     |

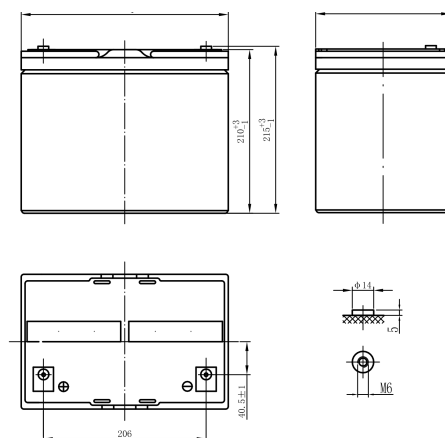
## Features

- Insulated intercell connections.
- Gas recombination efficiency up to 99%.
- High-rate discharge performance.
- Special welding process.
- Self-regulating valves; no water topping-up.
- Shelf life without recharge: 6 months at 20°C.
- Low internal pressure; special AGM separators.
- Optimized for 5–15 min; loads up to 1 hour.

## Applications

- Uninterruptible power supplies (UPS).
- Telecommunications backup power.
- Power and energy infrastructure.
- Security and fire alarm systems.
- Renewable-energy systems.
- Medical equipment.
- Emergency lighting.
- Data centres.

## Dimensions and terminal details / mm



Dimensions and tolerances govern; do not scale the drawing.

# Discharge performance

Fully charged battery / 25°C

## Constant-current discharge / A

| V/cell | 3m     | 5m     | 10m    | 15m    | 20m    | 25m    | 30m    | 45m   | 1h    |
|--------|--------|--------|--------|--------|--------|--------|--------|-------|-------|
| 1.60   | 397.00 | 332.00 | 242.00 | 183.00 | 153.00 | 130.00 | 116.00 | 87.80 | 69.10 |
| 1.65   | 386.00 | 319.00 | 231.00 | 174.00 | 148.00 | 127.00 | 113.00 | 85.20 | 67.30 |
| 1.67   | 382.00 | 314.00 | 227.00 | 170.00 | 146.00 | 126.00 | 112.00 | 84.20 | 66.60 |
| 1.70   | 372.00 | 305.00 | 216.00 | 164.00 | 140.00 | 122.00 | 109.00 | 82.20 | 64.20 |
| 1.75   | 348.00 | 272.00 | 198.00 | 154.00 | 131.00 | 113.00 | 102.00 | 79.10 | 62.20 |
| 1.80   | 318.00 | 249.00 | 183.00 | 144.00 | 124.00 | 108.00 | 98.00  | 73.60 | 57.90 |

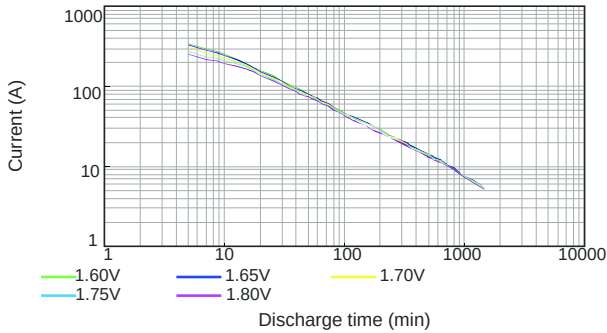
## Constant-power discharge / W per cell

| V/cell | 3m     | 5m     | 10m    | 15m    | 20m    | 25m    | 30m    | 45m    | 1h     |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1.60   | 810.00 | 640.00 | 432.00 | 327.00 | 281.00 | 241.00 | 213.00 | 158.00 | 124.00 |
| 1.65   | 754.00 | 598.00 | 420.00 | 322.00 | 273.00 | 235.00 | 209.00 | 155.00 | 122.00 |
| 1.67   | 732.00 | 581.00 | 415.00 | 320.00 | 270.00 | 232.00 | 207.00 | 154.00 | 121.00 |
| 1.70   | 673.00 | 551.00 | 386.00 | 301.00 | 263.00 | 228.00 | 204.00 | 152.00 | 119.00 |
| 1.75   | 643.00 | 500.00 | 364.00 | 288.00 | 249.00 | 216.00 | 196.00 | 150.00 | 116.00 |
| 1.80   | 583.00 | 465.00 | 340.00 | 273.00 | 238.00 | 210.00 | 188.00 | 142.00 | 111.00 |

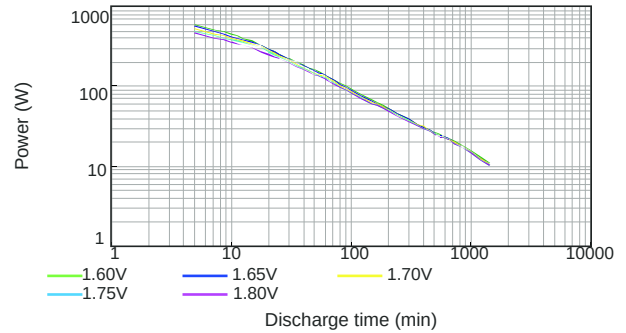
End voltage is per cell. Multiply W/cell by 6 for power per 12 V battery. Current is not divided by 6. Values are averages obtained within three charge/discharge cycles, not guaranteed minimums. Fully charge before testing. Product information may change without notice.

### Characteristic curves

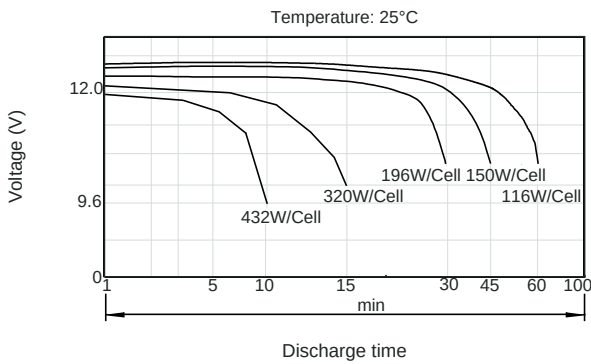
#### Constant-current discharge



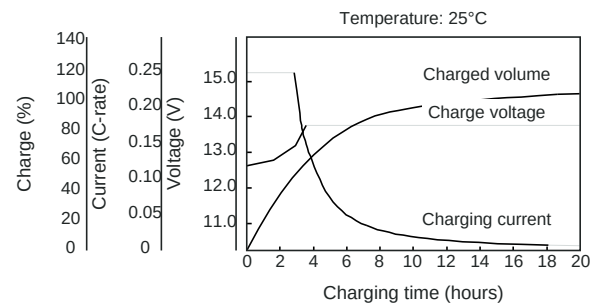
#### Constant-power discharge



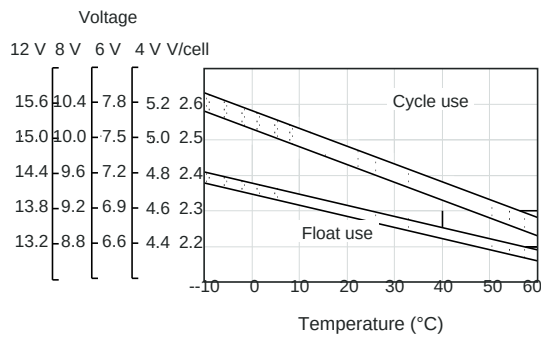
#### Discharge characteristics



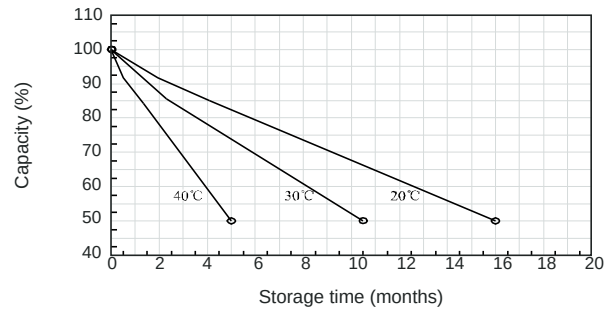
#### Charging characteristics



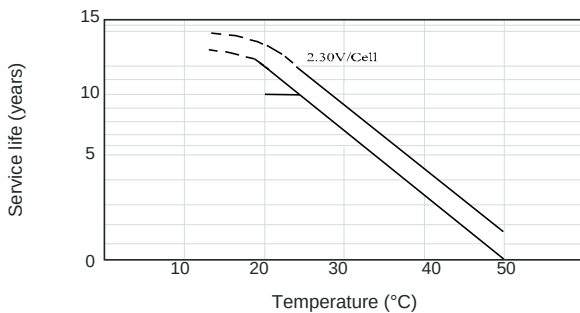
#### Charge voltage vs. temperature



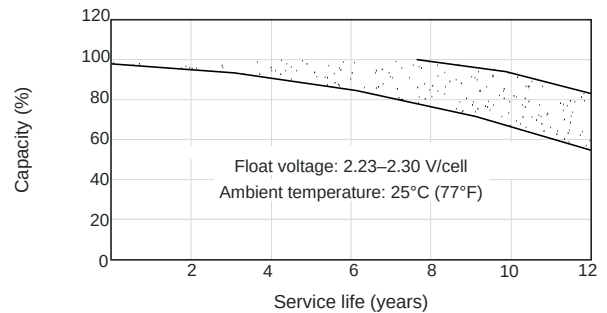
#### Self-discharge



#### Float life vs. temperature



#### Standby service life



Typical characteristic curves. Model-specific ratings are given on pages 1 and 3. Design life is not a warranty.